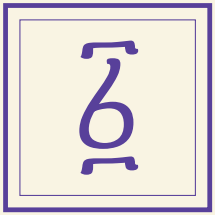
# Business cases

*Weekend 2: Representation learning*

Carlos Cotrini
*AI in Industry, ETH Zurich*
*Friday 13 November 2026* 𓂀𓋹𓂀𓆑



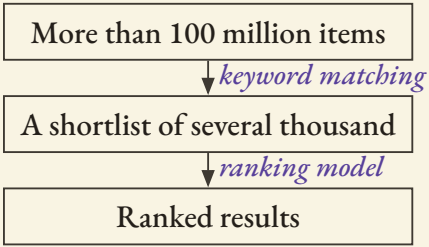
*Section 1*

# Etsy

# Etsy: a shortlist out of 100 million items

Etsy is a marketplace for handmade and vintage items.

Its shortlist used to come from keyword matching only, which fails when buyers and sellers use different words.

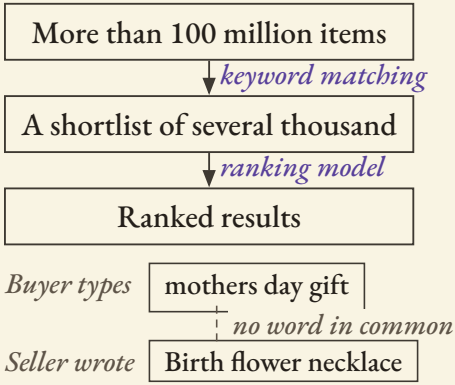



*Jha and others (Etsy), IEEE FMLDS 2024. The listing is invented.*

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*Jha and others (Etsy), IEEE FMLDS 2024. The listing is invented.*

# The training data: a month of search logs

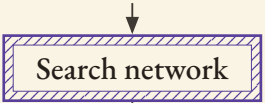
1 month: more than 150 million rows, from up to 30 million users.

| Search | Who searched | Item | Action |
|---|---|---|---|
| mothers day gift | Zurich, searched necklaces | Flower necklace | purchase |
| bridge photos | New York, clicked print shops | Bridge print | cart add |
| wool socks | Oslo, searched knitting | Knitted socks | cart add |

*The rows are invented. Jha and others (Etsy), IEEE FMLDS 2024.*

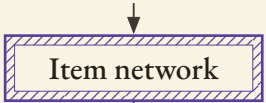
# Every search and every item becomes a list of numbers

The search words and who searched




*mothers day gift, in Zurich*

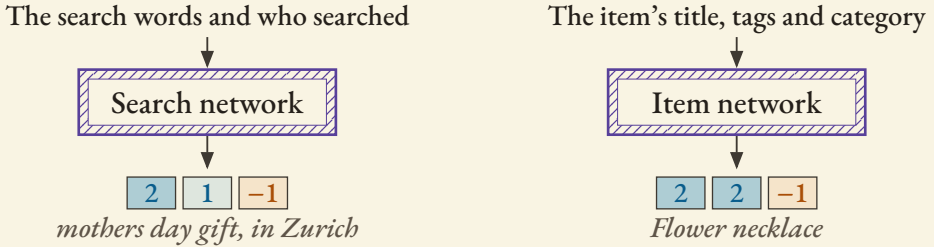
The item's title, tags and category




*Flower necklace*

*Invented numbers. Jha and others (Etsy), IEEE FMLDS 2024.*

# Every search and every item becomes a list of numbers




Training makes the inner product high for items bought or added to the cart after the search: here 7.

*Invented numbers. Jha and others (Etsy), IEEE FMLDS 2024.*

# A search arrives: sort by inner product

*mothers day gift* 2 1 −1

*Invented numbers.*

# A search arrives: sort by inner product

| | | | | |
|---|---|---|---|---|
| *mothers day gift* | 2 | 1 | −1 | *inner product* |
| | × | × | × | |
| Flower necklace | 2 | 2 | −1 | |
| | 4 | + 2 | + 1 | = 7 |

*Invented numbers.*

# A search arrives: sort by inner product

| | | | | inner product |
|---|---|---|---|---|
| *mothers day gift* | 2 | 1 | −1 | |
| | × | × | × | |
| Flower necklace | 2 | 2 | −1 | |
| | 4 | + 2 | + 1 | = 7 |
| Gift wrap | −1 | 0 | 2 | −4 |
| Flower necklace | 2 | 2 | −1 | 7 |
| Photo mug | 1 | 1 | 0 | 3 |
| Recipe board | 1 | 2 | 0 | 4 |

*Invented numbers.*

# A search arrives: sort by inner product

| | | | | inner product |
|---|---|---|---|---|
| *mothers day gift* | 2 | 1 | −1 | |
| | × | × | × | |
| Flower necklace | 2 | 2 | −1 | |
| | 4 | + 2 | + 1 | = 7 |
| Gift wrap | −1 | 0 | 2 | −4 |
| Flower necklace | 2 | 2 | −1 | 7 |
| Photo mug | 1 | 1 | 0 | 3 |
| Recipe board | 1 | 2 | 0 | 4 |

*Invented numbers.*

*Sorted: the top two join the shortlist*

| | |
|---|---|
| Flower necklace | 7 |
| Recipe board | 4 |
| Photo mug | 3 |
| Gift wrap | −4 |

# Cost and value

*Cost*

**72 hours** *on 4 GPUs*

**1 month** *of search logs*

*Value: A/B tests, relative change*

**+2.63%** *site conversion rate*

**+5.58%** *search purchase rate*

*Jha and others (Etsy), IEEE FMLDS 2024.*

# Cost and value

*Cost*

**72 hours** *on 4 GPUs*

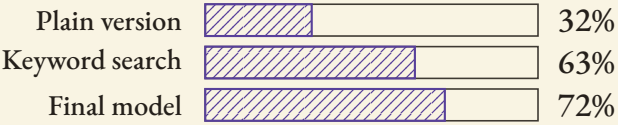
**1 month** *of search logs*

*Value: A/B tests, relative change*

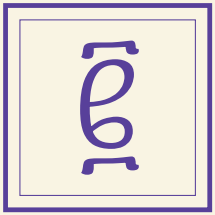
**+2.63%** *site conversion rate*

**+5.58%** *search purchase rate*

*Popular searches: share of the bought products found in the top 100*




*Jha and others (Etsy), IEEE FMLDS 2024.*

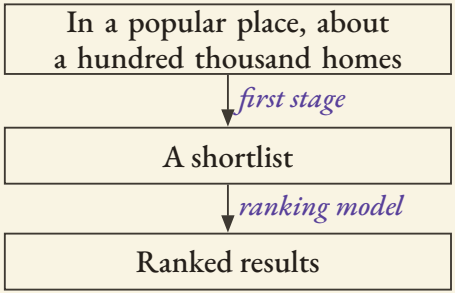


*Section 2*

# Airbnb

# Airbnb: the first stage of search

The first stage evaluates tens of thousands of homes.

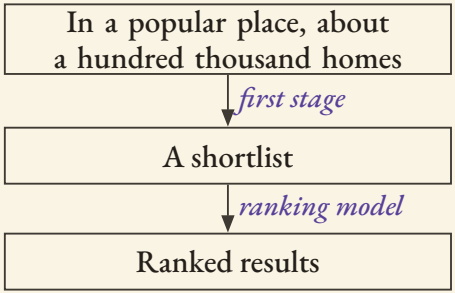



*Abdool and others (Airbnb), arXiv preprint 2026; Airbnb's blog 2025.*

# Airbnb: the first stage of search

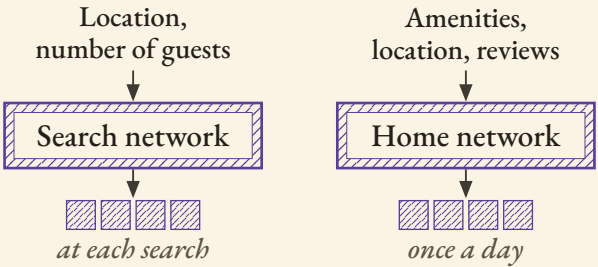
The first stage evaluates tens of thousands of homes.

It used to be only a simple linear model that scored every home once a day, whatever the search.



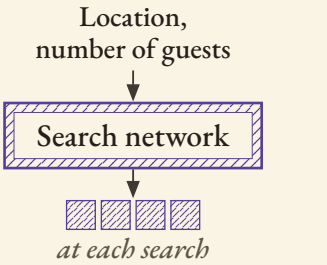

*Abdool and others (Airbnb), arXiv preprint 2026; Airbnb's blog 2025.*

# Two networks and the nearest homes

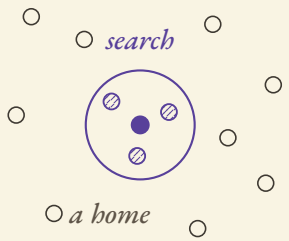



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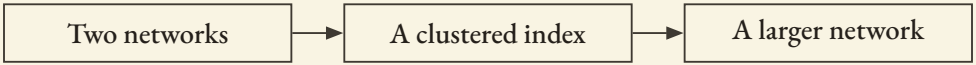



*The nearest homes join the shortlist*



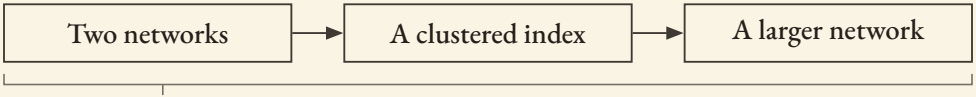

*Abdool and others (Airbnb), arXiv preprint 2026; Airbnb's blog 2025.*

# The value: three A/B tests




*Abdool and others (Airbnb), arXiv preprint 2026; Airbnb's blog 2025.*

# The value: three A/B tests



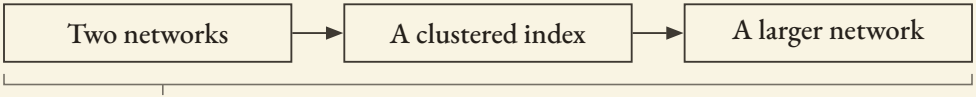

**+0.31%**

*conversion, all three, relative change*

Airbnb: “a large portion of the team’s annual goal”.

*Abdool and others (Airbnb), arXiv preprint 2026; Airbnb’s blog 2025.*

# The value: three A/B tests




**+0.31%**

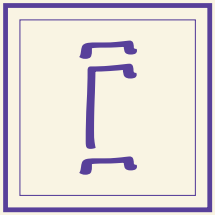
*conversion, all three, relative change*

Airbnb: “a large portion of the team’s annual goal”.

**+2.3%** *bookings from promotional emails*

**−16%** *compute with the clustered index*

*Abdool and others (Airbnb), arXiv preprint 2026; Airbnb’s blog 2025.*



*Section 3*

# Spotify

# Spotify's audiobooks: little data to learn from

When Spotify launched audiobooks it had little data to learn from.

Each audiobook had to be bought before it could be streamed, and about a quarter of the audiobook listeners made three quarters of the listening hours.

*De Nadai and others (Spotify), WWW 2024 Companion.*

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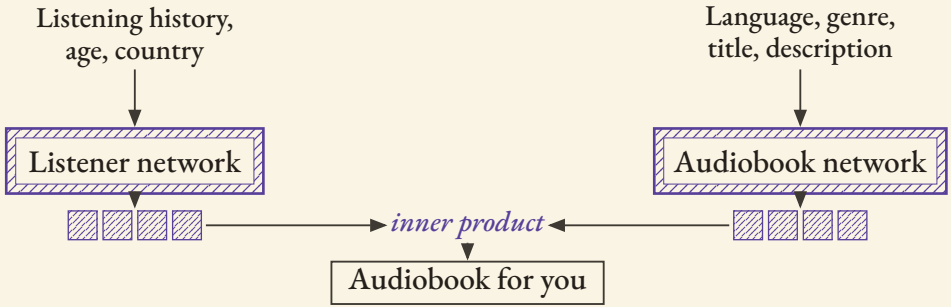
**Over 70%**

of the first audiobook listeners already listened to podcasts.

*De Nadai and others (Spotify), WWW 2024 Companion.*

# 128 numbers for every listener and every audiobook

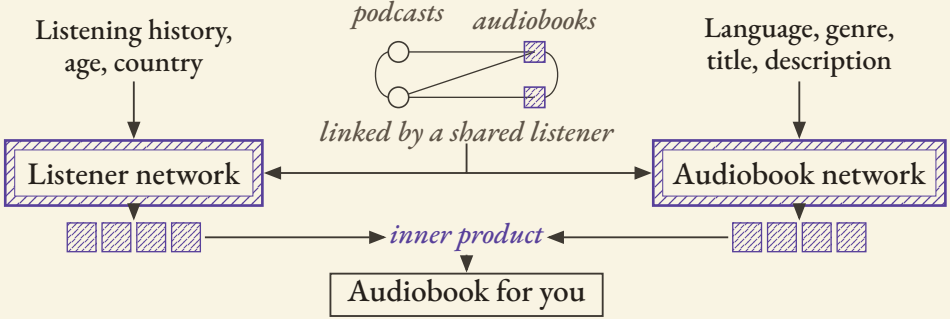
The matchmaker learned the numbers. Here two networks compute them.




*De Nadai and others (Spotify), WWW 2024 Companion.*

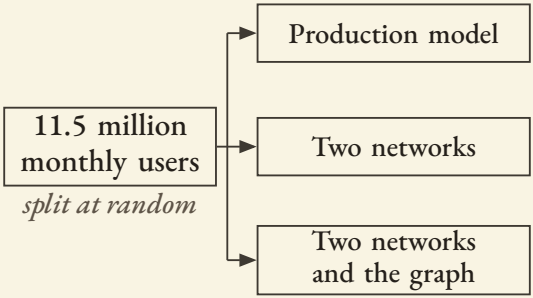
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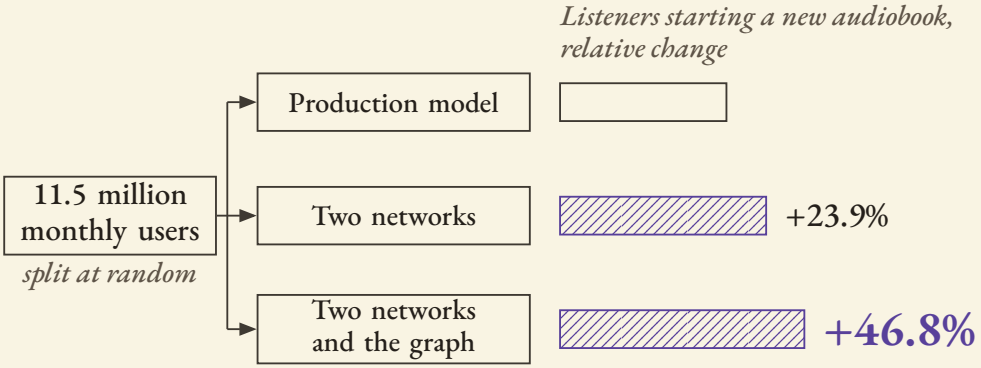

*De Nadai and others (Spotify), WWW 2024 Companion.*

# The value: an A/B test

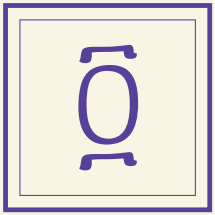



*De Nadai and others (Spotify), WWW 2024 Companion.*

# The value: an A/B test




*De Nadai and others (Spotify), WWW 2024 Companion.*



*Section 4*

# Spotify’s home page

# Spotify's home page: music or podcasts?

The home page shows shelves of music and podcasts. A common rule gives users the mix of their past listening.

But tastes can change with the hour, the day of the week, even the device.

*Feijer and others (Spotify), CONSEQUENCES workshop at RecSys 2025.*

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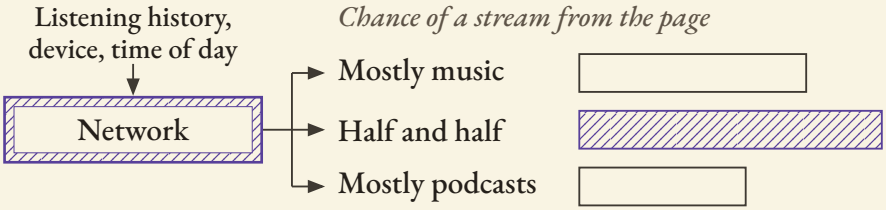
**Over 600 million**

users of the home page

*Feijer and others (Spotify), CONSEQUENCES workshop at RecSys 2025.*

# Epsilon-greedy on the home page

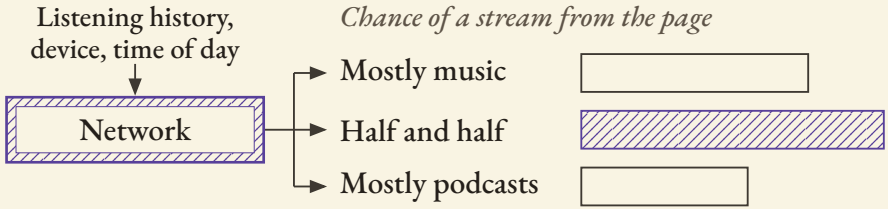
The mirror counted taps. Here a network estimates the chances.




*Invented mixes and estimates. Feijer and others (Spotify), CONSEQUENCES workshop at RecSys 2025.*

# Epsilon-greedy on the home page

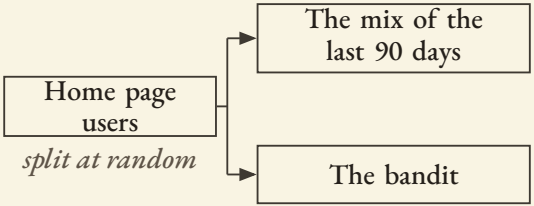
The mirror counted taps. Here a network estimates the chances.




**98.5%** of the pages: the highest estimate.
**1.5%**: a random mix near the usual one.

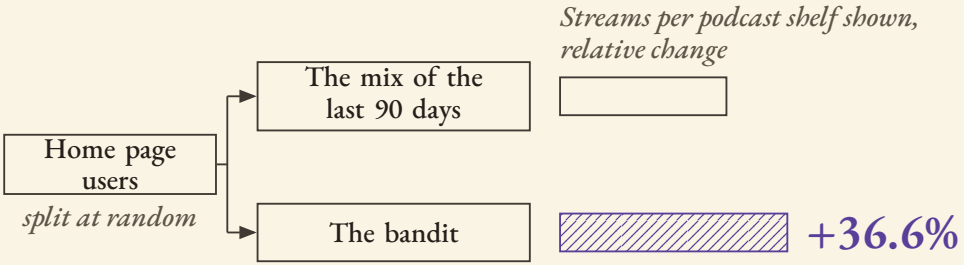
*Invented mixes and estimates. Feijer and others (Spotify), CONSEQUENCES workshop at RecSys 2025.*

# The value: an A/B test



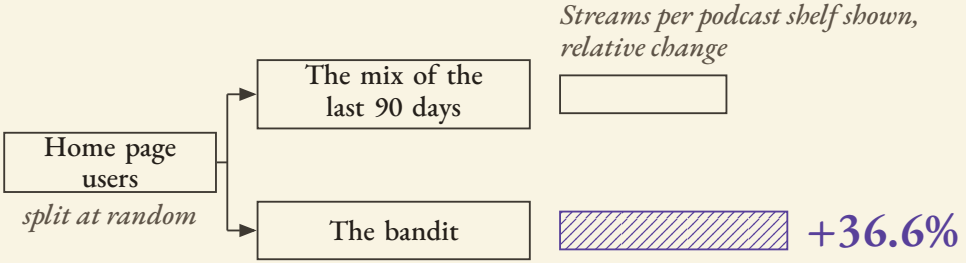

*Feijer and others (Spotify), CONSEQUENCES workshop at RecSys 2025.*

# The value: an A/B test




*Feijer and others (Spotify), CONSEQUENCES workshop at RecSys 2025.*

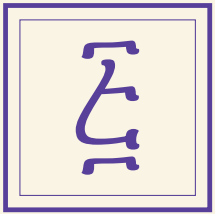
# The value: an A/B test




In production as of March 2025.

+1.28% *minutes listened on Spotify, relative change*

*Feijer and others (Spotify), CONSEQUENCES workshop at RecSys 2025.*

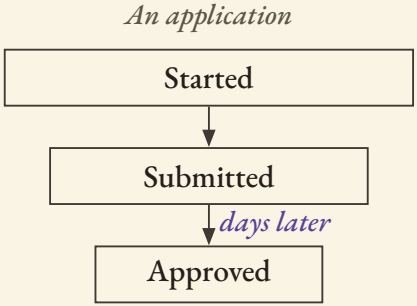


*Section 5*

# Amazon Payments

# Amazon Payments: which page for which visitor?

A page invites customers to apply for an Amazon Payments product. It shows one image and one tagline, from many possible pairs.

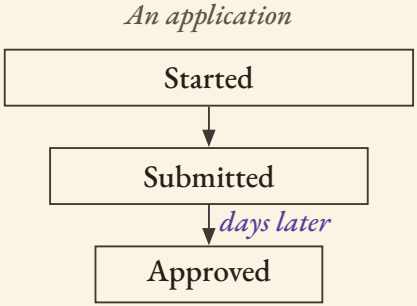



*John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.*

# Amazon Payments: which page for which visitor?

A page invites customers to apply for an Amazon Payments product. It shows one image and one tagline, from many possible pairs.

A page that wins the first step can lose the last.




*John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.*

# One UCB score for each step

Each score is an estimate plus a bonus, as in the mirror.

| *For one visitor* | *Started* | *Submitted* | *Approved* |
|---|---|---|---|
| Pair A | 6 | 3 | 1 |
| Pair B | 4 | 4 | 4 |
| Pair C | 3 | 2 | 3 |

The first step alone would choose pair A.

*Invented scores. John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.*

# One UCB score for each step

Each score is an estimate plus a bonus, as in the mirror.

| *For one visitor* | *Started* | *Submitted* | *Approved* | *Sum* |
|---|---|---|---|---|
| Pair A | 6 | 3 | 1 | 10 |
| Pair B | 4 | 4 | 4 | 12 |
| Pair C | 3 | 2 | 3 | 8 |

The first step alone would choose pair A.
Amazon shows the highest sum: pair B.

*Invented scores. John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.*

# The value: an A/B test of 7 weeks

The bandit against the fixed page:

One group of customers



Approvals rose by a high single-digit percentage.
*relative change so far, not reported as significant*

*John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.*

# The value: an A/B test of 7 weeks

The bandit against the fixed page:

| | | |
|---|---|---|
| One group of customers |  | Approvals rose by a high single-digit percentage. *relative change so far, not reported as significant* |
| Another group |  | No pair it tried beat the fixed page. Approvals fell. *statistically significant* |

*John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.*

# The value: an A/B test of 7 weeks

The bandit against the fixed page:

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Amazon: “Thorough exploration can confirm that a content pool contains no winner.”

*John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.*

6

*Section 6*

# LinkedIn's ads

# LinkedIn's ads: the chance of a click

The chance that a member clicks an ad enters the ad auction, and the price depends on it.

Advertisers choose which clicks they pay for: some count likes or comments, others only visits to their website.

*Borisyuk and others (LinkedIn), KDD 2024.*

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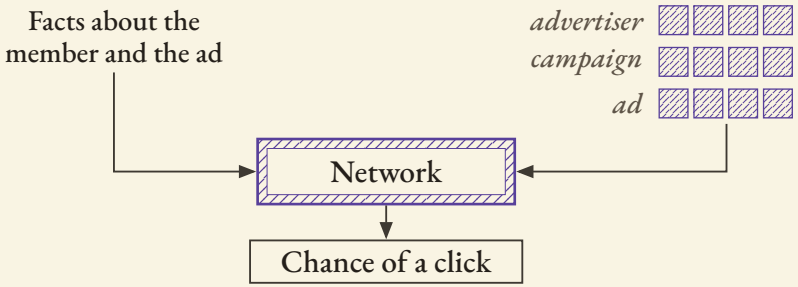
**Over 1 billion**

members

*Borisyuk and others (LinkedIn), KDD 2024.*

# Numbers for every advertiser, campaign and ad

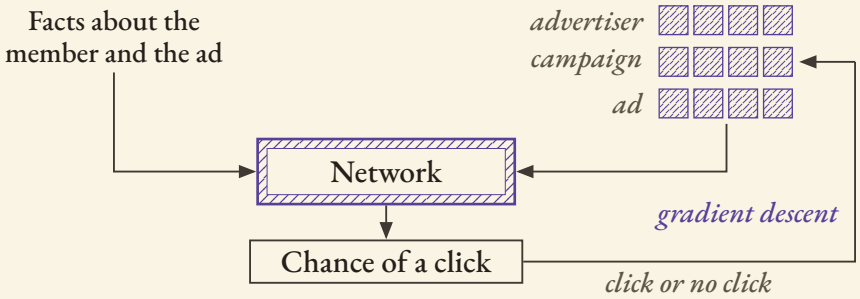
The matchmaker learned lists from ratings; here, from clicks.




*Borisyuk and others (LinkedIn), KDD 2024.*

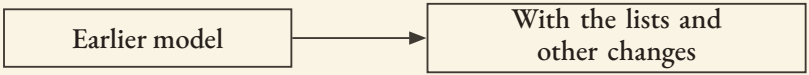
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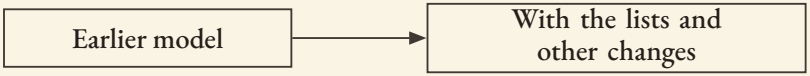

*Borisyuk and others (LinkedIn), KDD 2024.*

# The value: A/B tests




*Borisyuk and others (LinkedIn), KDD 2024.*

# The value: A/B tests



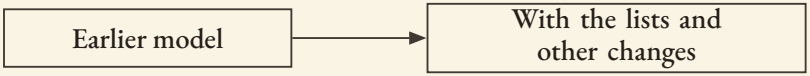

**+4.3%**

*click rate on ads, relative change*

LinkedIn put the new model into production.

*Borisyuk and others (LinkedIn), KDD 2024.*

# The value: A/B tests




**+4.3%**

*click rate on ads, relative change*

LinkedIn put the new model into production.

**−96%**

*training time in an offline test, when each new model builds on the earlier ones and learns from the newest half day of data*

*Borisyuk and others (LinkedIn), KDD 2024.*